

A deterministic lower bound for the binomial triple projection

Research proof candidate; independent review pending

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Let $\rho_1, \dots, \rho_m$ be arbitrary permutations of $[n]$, with no randomness or independence assumption. On distinct labels put

$$\delta_i(a, b, c) = \frac{1}{2} - \frac{3}{2} \mathbf{1}\{\rho_i(b) < \rho_i(a), \rho_i(b) < \rho_i(c)\}.$$

Write $d = n - 3$, $p_i = (i - 1)/(m - 1)$, and

$$w_i(a) = \binom{d}{a} p_i^a (1 - p_i)^{d-a}, \quad h = \binom{n}{3} m^{-3} (1 - 1/m)^{n-3}.$$

The triple contribution in the palindrome expansion is the following function on target permutations:

$$S(\sigma) = h \sum_{a=0}^d \sum_{i=1}^m w_i(a) \delta_i(\sigma_{a+1}, \sigma_{a+2}, \sigma_{a+3}).$$

The terminology “projection” refers to the decomposition for the independent random model. The displayed function itself makes sense for every deterministic block sequence.

Theorem 1. *There are absolute constants $c_0, c_1 > 0$ and n_0 such that, for $n \geq n_0$ and*

$$n \leq m \leq c_0 n^{3/2},$$

every choice of the block permutations satisfies

$$\max_{\sigma \in S_n} S(\sigma) - \min_{\sigma \in S_n} S(\sigma) \geq c_1 \frac{n^{13/4}}{m^{5/2}}.$$

The range is witnessed by independent middle-pair swaps in disjoint six-label windows in the central third of the target permutation.

Scope. This theorem concerns the displayed linear contribution. It is not a lower bound for the complete ordering error of arbitrary deterministic palindrome sequences. The other terms can depend strongly on the block sequence; the probabilistic remainder bounds for independent blocks cannot be used for a deterministic sequence.

Lemma 2 (Central binomial weights). *For d sufficiently large, $d/4 \leq a \leq 3d/4$, and $m \geq d + 3$, put $Z_a = \sum_i w_i(a)$. There are absolute constants $C, c > 0$ such that*

$$c \frac{m}{d} \leq Z_a \leq C \frac{m}{d}, \quad c \frac{m}{d^{3/2}} \leq \sum_i w_i(a)^2 \leq C \frac{m}{d^{3/2}},$$

$$\sum_i |w_i(a+r) - w_i(a)| \leq C \frac{m}{d^{3/2}} \quad (|r| \leq 3),$$

provided the shifted indices also lie in $[d/4, 3d/4]$. In particular, the normalized weights $\mu_i = w_i(a)/Z_a$ satisfy $\sum_i \mu_i^2 \asymp \sqrt{d}/m$.

Proof. Let $f_a(p) = \binom{d}{a} p^a (1-p)^{d-a}$. It is unimodal, has integral $1/(d+1)$, and, by Stirling's bounds, maximum at most $C/\sqrt{d}$ uniformly over central a . For a function f of bounded variation, its sum on the grid of mesh $1/(m-1)$ differs from $(m-1) \int f$ by at most its total variation plus endpoint terms. Consequently the error for f_a is $O(d^{-1/2})$, proving the estimates for Z_a . For the squared weights, the beta integral and the same Stirling bounds give

$$\int_0^1 f_a(p)^2 dp = \frac{\binom{d}{a}^2}{(2d+1)\binom{2d}{2a}} \asymp d^{-3/2}.$$

The squared function is unimodal with maximum $O(d^{-1})$, so its grid error is $O(d^{-1})$, smaller than its main term for $m \geq d + 3$ and sufficiently large d . This proves the squared-weight estimate.

For neighboring indices, let $g(p) = \binom{d+1}{a+1} p^{a+1} (1-p)^{d-a}$. Direct differentiation gives

$$g'(p) = (d+1)(f_a(p) - f_{a+1}(p)).$$

Thus $\int |f_a - f_{a+1}| = 2 \max g/(d+1) = O(d^{-3/2})$. The total variation of $|f_a - f_{a+1}|$ is at most the sum of the variations of f_a, f_{a+1} , hence is $O(d^{-1/2})$. The grid estimate therefore gives $O(md^{-3/2} + d^{-1/2}) = O(md^{-3/2})$. Telescoping proves the claim for all the stated shifts. $\square$

Proof of Theorem 1. Use the weighted six-window theorem in the companion note *Weighted collision bounds for triple minima and six-window swaps*: for a weighted permutation family on a set of N labels with $\kappa = \sum_i \mu_i^2$ and $N \geq C_*/\kappa$, some six distinct labels have a middle-pair swap changing the sum of their four consecutive triple scores by at least $c_*\sqrt{\kappa}$. Its constants are absolute, and its conclusion is preserved on restriction to a label subset.

Fix $J = \lfloor n/40 \rfloor$ disjoint six-position windows in the central third of the target order; a harmless increase in n_0 ensures that this is possible. Fill these windows successively with labels, always using labels not used in earlier windows. At every step at least $n - 6J \geq n/2$ labels are available. If a is the start of the first of the four affected triples in the current window, all starts $a, a+1, a+2, a+3$ are central.

Take the distribution $\mu_i = w_i(a)/Z_a$. By Lemma 2, $\kappa \asymp \sqrt{n}/m$. We can choose c_0 sufficiently small that $n/2 \geq C_*/\kappa$. Apply the weighted six-window theorem on the unused labels. It supplies six labels and a swap with absolute normalized score difference at least $cn^{1/4}/\sqrt{m}$.

Replace each constant weight $w_i(a)$ by the actual weight $w_i(a+r)$ in its corresponding triple, for $r = 0, 1, 2, 3$. Lemma 2 bounds the resulting normalized change by

$$\frac{2}{Z_a} \sum_{r=0}^3 \sum_i |w_i(a+r) - w_i(a)| \leq \frac{C}{\sqrt{n}}.$$

After decreasing c_0 once more, this error is less than half the supplied swap difference. Thus the actual change in S from this window is at least

$$chZ_a \frac{n^{1/4}}{\sqrt{m}} \geq ch \frac{m n^{1/4}}{n \sqrt{m}}.$$

Repeat for all windows and put the remaining labels arbitrarily in the remaining positions. The two exterior labels on each side of each swapped pair stay fixed, so all affected triples lie inside their own window. The effects of the J binary choices are exactly additive. Hence the range is at least the sum of the J absolute changes. Using $J \geq cn$ and $h \geq cn^3/m^3$ proves

$$\text{range}(S) \geq cn \frac{n^3}{m^3} \frac{m n^{1/4}}{n \sqrt{m}} = c \frac{n^{13/4}}{m^{5/2}},$$

as required. □